

Sleep Quality Analysis Using Machine Learning and Explainable Artificial Intelligence

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Abstract—Sleep quality is an important indicator of physical health, mental stability, and daily productivity. This project presents a machine learning based sleep quality analysis system using a sleep health and lifestyle dataset. The work includes data preprocessing, feature engineering, clustering analysis, supervised classification, hyperparameter tuning, feature importance analysis, and a recommendation system. The original sleep quality score was converted into a binary target variable, where sleep quality greater than or equal to 7 was labeled as Good and otherwise labeled as Poor. K-Means clustering was applied to discover hidden sleep behavior groups, while several machine learning models such as Naive Bayes, Support Vector Machine, K-Nearest Neighbors, Decision Tree, Random Forest, Logistic Regression, Gradient Boosting, AdaBoost, Extra Trees, Hist Gradient Boosting, and Voting Classifier were trained and evaluated. Experimental results showed that Random Forest, Decision Tree, Logistic Regression, AdaBoost, and several advanced ensemble models achieved very high classification performance. Feature importance analysis indicated that sleep stress ratio, heart-rate stress, stress level, sleep duration, and heart rate were among the most influential factors for sleep quality prediction.

Index Terms—Sleep Quality, Machine Learning, Random Forest, K-Means Clustering, Feature Engineering, Classification, Explainable AI, Health Data Analysis

I. INTRODUCTION

Sleep is one of the most essential biological processes for human health. Proper sleep supports memory consolidation, emotional balance, immune system function, physical recovery, and cognitive performance. In contrast, poor sleep quality can lead to fatigue, stress, lack of concentration, reduced productivity, and long-term health risks. Therefore, analyzing sleep behavior is an important task in modern health informatics.

With the growth of health-related datasets, machine learning has become a useful approach for identifying patterns in lifestyle and medical data. Machine learning algorithms can learn relationships between input features and target outcomes. In this project, sleep quality is predicted using lifestyle and health-related attributes such as age, gender, sleep duration, physical activity level, stress level, heart rate, daily steps, blood pressure, body mass index category, and occupation.

The main purpose of this project is not only to classify sleep quality but also to understand the factors that influence it. For this reason, the project includes both predictive modeling and interpretability. In addition, clustering analysis was performed to discover natural groups of individuals based on sleep-related

characteristics. Finally, a rule-based recommendation system was developed to provide personalized sleep improvement suggestions.

II. OBJECTIVES

The major objectives of this project are as follows:

- To load and analyze the sleep health and lifestyle dataset.
- To preprocess the dataset by handling missing values and categorical variables.
- To convert the numerical sleep quality score into a binary classification target.
- To engineer additional meaningful features related to sleep, stress, activity, and heart rate.
- To apply K-Means clustering for discovering hidden sleep pattern groups.
- To train and compare multiple machine learning classification models.
- To tune the Random Forest model using Grid Search.
- To analyze important features affecting sleep quality.
- To generate sleep improvement recommendations based on health and lifestyle features.

III. DATASET DESCRIPTION

The dataset used in this project is the Sleep Health and Lifestyle dataset. It contains personal, lifestyle, and health-related information. The dataset initially contained 374 records. During preprocessing, the column named Sleep Disorder was removed because it contained excessive missing values. After removing rows with remaining missing values in the basic pipeline, 373 records and 13 columns were retained.

The important features used in this project include Age, Gender, Occupation, Sleep Duration, Quality of Sleep, Physical Activity Level, Stress Level, BMI Category, Blood Pressure, Heart Rate, Daily Steps, and Sleep Disorder. The target variable was created from the original Quality of Sleep feature. If the quality score was greater than or equal to 7, the sample was labeled as Good. Otherwise, it was labeled as Poor.

IV. DATA PREPROCESSING

Data preprocessing is an important step before applying machine learning algorithms. Raw datasets may contain missing values, categorical variables, irrelevant columns, and features with different numerical ranges. If these issues are not handled properly, model performance can decrease.

TABLE I
IMPORTANT DATASET ATTRIBUTES

Attribute	Description
Age	Age of the person
Gender	Male or female category
Occupation	Professional category of the person
Sleep Duration	Total sleeping time in hours
Quality of Sleep	Numerical sleep quality score
Physical Activity Level	Daily physical activity measurement
Stress Level	Stress score of the person
BMI Category	Body mass index category
Blood Pressure	Systolic and diastolic blood pressure
Heart Rate	Heart beats per minute
Daily Steps	Number of daily walking steps

A. Missing Value Handling

The Sleep Disorder column was dropped because it contained more than 30 percent missing values. The threshold used for removing high-missing-value columns was 0.30. After removing this column, remaining rows containing missing values were removed. In the basic pipeline, the cleaned dataset shape became 373 rows and 13 columns.

B. Target Variable Creation

The original Quality of Sleep feature was numerical. For classification, it was transformed into a binary categorical target using the following rule:

$$Y = \begin{cases} \text{Good}, & \text{if Quality of Sleep} \geq 7 \\ \text{Poor}, & \text{if Quality of Sleep} < 7 \end{cases} \quad (1)$$

This target variable was named Quality_of_Sleep_Cat.

C. Categorical Encoding

Machine learning algorithms require numerical input. Therefore, categorical columns such as Gender, Occupation, BMI Category, Daily Steps, and Blood Pressure were converted using one-hot encoding. One-hot encoding creates binary columns for categorical values and helps models process non-numeric information.

D. Feature Scaling

For clustering and some classification models, feature scaling was applied. Standardization transforms a feature using the following equation:

$$X' = \frac{X - \mu}{\sigma} \quad (2)$$

where X is the original value, μ is the mean, and σ is the standard deviation.

V. FEATURE ENGINEERING

Feature engineering was used to create additional meaningful variables from existing attributes. This helps machine learning models learn stronger patterns.

A. Sleep Efficiency

An age-based sleep efficiency feature was created. First, an age factor was assigned based on age group. For age less than or equal to 35, the factor was 7.5. For age between 36 and 50, the factor was 7.0. For age greater than 50, the factor was 6.5. Sleep efficiency was calculated as:

$$\text{Sleep Efficiency} = \frac{\text{Sleep Duration}}{\text{Stress Level} \times \text{Age Factor}} \quad (3)$$

B. Advanced Features

In the advanced pipeline, several new features were created:

- Sleep_Stress_Ratio
- Steps_per_Hour_Sleep
- Activity_Stress_Index
- HeartRate_Stress
- Systolic_BP
- Diastolic_BP
- BMI_Level

The Sleep_Stress_Ratio was computed as:

$$\text{Sleep_Stress_Ratio} = \frac{\text{Sleep Duration}}{\text{Stress Level} + 1} \quad (4)$$

The Steps_per_Hour_Sleep feature was computed as:

$$\text{Steps_per_Hour_Sleep} = \frac{\text{Daily Steps}}{\text{Sleep Duration} + 1} \quad (5)$$

The Activity_Stress_Index was computed as:

$$\text{Activity_Stress_Index} = \frac{\text{Physical Activity Level}}{\text{Stress Level} + 1} \quad (6)$$

The HeartRate_Stress feature was computed as:

$$\text{HeartRate_Stress} = \text{Heart Rate} \times \text{Stress Level} \quad (7)$$

After advanced feature engineering and cleaning, the advanced dataframe contained 154 records and 21 columns.

VI. CLUSTERING ANALYSIS

K-Means clustering was applied to discover hidden sleep behavior groups. The clustering features were Sleep Duration, Physical Activity Level, Stress Level, Heart Rate, Daily Steps, and Sleep Efficiency. These features were scaled before clustering.

K-Means minimizes the within-cluster sum of squared distances:

$$J = \sum_{i=1}^k \sum_{x \in C_i} ||x - \mu_i||^2 \quad (8)$$

where k is the number of clusters, C_i is the i -th cluster, and μ_i is the centroid of that cluster.

The elbow method was used to select the number of clusters. In this project, $k = 3$ was selected. The clustering result produced three sleep behavior groups.

Cluster 0 represents individuals with better sleep duration, higher physical activity, lower stress level, lower heart rate, and higher sleep efficiency. Cluster 1 contains individuals with lower sleep duration and high stress level. Cluster 2 contains individuals with moderate sleep duration but lower daily steps and higher heart rate.

TABLE II
K-MEANS CLUSTER SUMMARY

Feature	Cluster 0	Cluster 1	Cluster 2
Sleep Duration	7.60	6.27	6.64
Physical Activity	62.88	53.99	46.50
Stress Level	4.27	7.44	6.41
Heart Rate	67.77	72.88	81.32
Daily Steps	6973.50	7035.90	3977.27
Sleep Efficiency	0.278	0.121	0.160

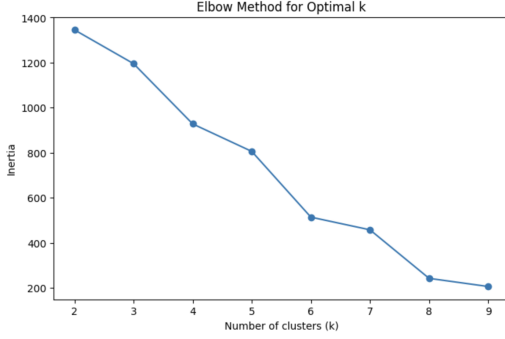


Fig. 1. Elbow method for selecting optimal number of clusters.

VII. MACHINE LEARNING MODELS

Several supervised learning algorithms were used for sleep quality classification.

A. Naive Bayes

Naive Bayes is a probabilistic classifier based on Bayes theorem. It assumes that features are conditionally independent. Although this assumption may not always be true, Naive Bayes often performs well on classification tasks.

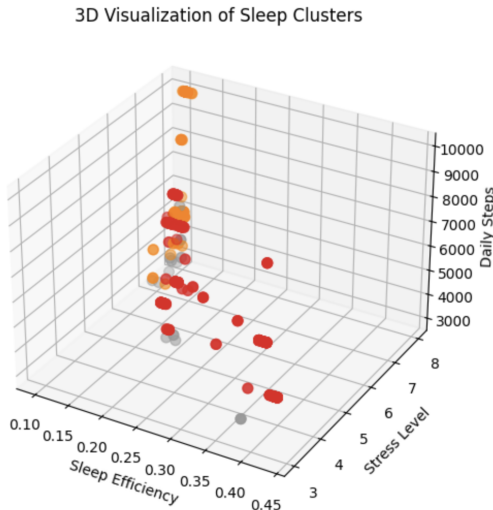


Fig. 2. Three-dimensional visualization of sleep clusters.

B. Support Vector Machine

Support Vector Machine attempts to find an optimal separating hyperplane between classes. In the basic experiment, SVM produced lower performance because it predicted only the Good class for many samples. In the advanced pipeline, SVM with RBF kernel performed much better.

C. K-Nearest Neighbors

K-Nearest Neighbors classifies a sample based on the majority class among its nearest neighbors. It is simple and effective but sensitive to feature scaling and distance measurement.

D. Decision Tree

Decision Tree creates a tree-like structure by splitting data based on feature values. It is easy to interpret, but it may overfit the training data if not controlled.

E. Random Forest

Random Forest is an ensemble method that combines multiple decision trees. It reduces overfitting and improves generalization by using bootstrap sampling and random feature selection.

F. Logistic Regression

Logistic Regression is a linear classification algorithm. It estimates the probability of a class using the sigmoid function:

$$P(y = 1) = \frac{1}{1 + e^{-z}} \quad (9)$$

G. Gradient Boosting and AdaBoost

Boosting algorithms build models sequentially. Each new model attempts to correct the errors of previous models. Gradient Boosting and AdaBoost were used to improve predictive performance.

H. Advanced Ensemble Models

The advanced section used Random Forest, Extra Trees, Gradient Boosting, Hist Gradient Boosting, SVM RBF, Logistic Regression, and Voting Classifier. Voting Classifier combined Random Forest, Extra Trees, and Logistic Regression using soft voting.

VIII. EXPERIMENTAL SETUP

The dataset was divided into training and testing sets. In the basic pipeline, the test set contained 112 samples. In the advanced pipeline, after feature engineering and cleaning, the dataset was split into 107 training samples and 47 testing samples. The advanced dataset contained 40 encoded features after one-hot encoding.

The evaluation metrics used in this project were accuracy, precision, recall, and F1-score.

Accuracy is defined as:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (10)$$

Precision is defined as:

$$Precision = \frac{TP}{TP + FP} \quad (11)$$

Recall is defined as:

$$Recall = \frac{TP}{TP + FN} \quad (12)$$

F1-score is defined as:

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (13)$$

IX. BASIC MODEL RESULTS

The basic model comparison showed that several models achieved strong performance. Naive Bayes achieved 98.21 percent accuracy. K-Nearest Neighbors achieved 90.18 percent accuracy. Decision Tree, Random Forest, Logistic Regression, and AdaBoost achieved 100 percent accuracy on the test set. Gradient Boosting achieved 99.11 percent accuracy. SVM achieved 68.75 percent accuracy because it failed to correctly predict the Poor class.

TABLE III
BASIC MODEL ACCURACY COMPARISON

Model	Accuracy
Naive Bayes	0.9821
SVM	0.6875
K-Nearest Neighbors	0.9018
Decision Tree	1.0000
Random Forest	1.0000
Logistic Regression	1.0000
Gradient Boosting	0.9911
AdaBoost	1.0000

TABLE IV
CONFUSION MATRIX EXAMPLES FROM BASIC PIPELINE

Model	Good-Good	Good-Poor	Poor-Good	Poor-Poor
Naive Bayes	75	2	0	35
SVM	77	0	35	0
KNN	71	6	5	30
Random Forest	77	0	0	35
Gradient Boosting	77	0	1	34

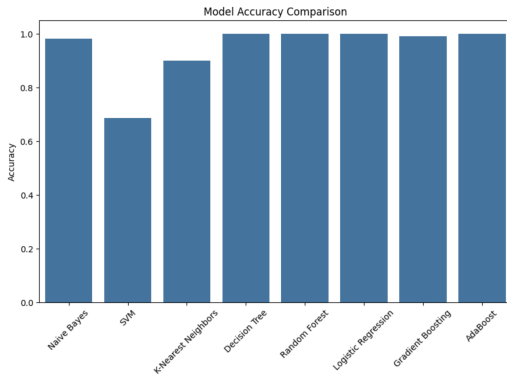


Fig. 3. Accuracy comparison of basic machine learning models.

X. HYPERPARAMETER TUNING

Random Forest was tuned using Grid Search with 5-fold cross-validation. The parameter grid included number of estimators, maximum depth, and minimum samples split. The tested values were:

- n_estimators: 100, 200, 300
- max_depth: None, 10, 20, 30
- min_samples_split: 2, 5, 10

The best parameters obtained from Grid Search were:

- max_depth: None
- min_samples_split: 2
- n_estimators: 100

The tuned Random Forest achieved 100 percent accuracy on the test set with perfect precision, recall, and F1-score for both Good and Poor classes.

XI. ADVANCED MODEL RESULTS

The advanced model comparison was performed after advanced feature engineering. The advanced dataframe contained 154 records and 21 columns. After one-hot encoding, the feature matrix contained 40 features. The training set contained 107 samples and the test set contained 47 samples.

TABLE V
ADVANCED MODEL ACCURACY COMPARISON

Model	Accuracy
Random Forest	1.0000
Extra Trees	1.0000
Gradient Boosting	1.0000
Hist Gradient Boosting	1.0000
SVM RBF	0.9787
Logistic Regression	1.0000
Voting Classifier	1.0000

The advanced results show that most ensemble models achieved perfect accuracy on the test set. SVM RBF achieved 97.87 percent accuracy, with precision and recall close to 0.98. These results indicate that the engineered features improved separability between Good and Poor sleep quality classes.

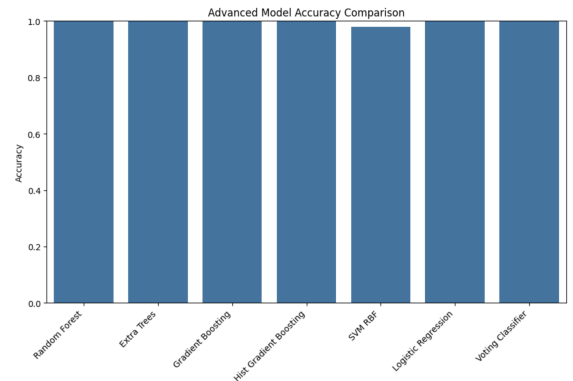


Fig. 4. Accuracy comparison of advanced machine learning models.

XII. CROSS-VALIDATION ANALYSIS

Cross-validation was applied to evaluate model stability. The 5-fold cross-validation results showed that Logistic Regression achieved a mean cross-validation accuracy of 1.0000 with zero standard deviation. Random Forest and Hist Gradient Boosting achieved mean accuracy of 0.9742 with standard deviation of 0.0241. Extra Trees achieved mean accuracy of 0.9355, while Gradient Boosting achieved 0.8839. SVM RBF achieved 0.6710 mean accuracy with high standard deviation.

TABLE VI
5-FOLD CROSS-VALIDATION RESULTS

Model	Mean Accuracy	Std. Dev.
Random Forest	0.9742	0.0241
Extra Trees	0.9355	0.1136
Gradient Boosting	0.8839	0.1750
Hist Gradient Boosting	0.9742	0.0241
SVM RBF	0.6710	0.2710
Logistic Regression	1.0000	0.0000

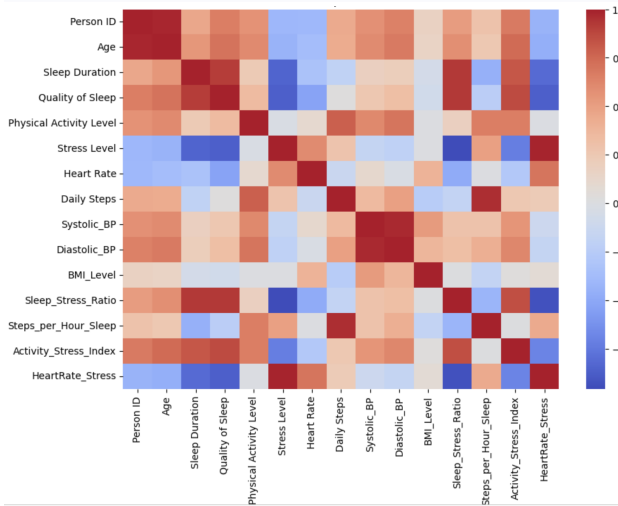


Fig. 5. Five-fold cross-validation accuracy comparison.

XIII. FEATURE IMPORTANCE ANALYSIS

Feature importance was calculated using a Random Forest model with 300 estimators. The top features showed that stress-related and sleep-duration-related variables had strong influence on sleep quality prediction.

The most important feature was Sleep_Stress_Ratio. This indicates that the relationship between sleep duration and stress level is highly relevant for sleep quality. HeartRate_Stress was also highly important, showing that the combined effect of heart rate and stress can help distinguish between Good and Poor sleep categories.

XIV. CONFUSION MATRIX OF BEST MODEL

The best advanced model selected from the accuracy dictionary was Random Forest. It achieved 100 percent accuracy on the advanced test set. The confusion matrix showed that

TABLE VII
TOP IMPORTANT FEATURES

Feature	Importance
Sleep_Stress_Ratio	0.1741
HeartRate_Stress	0.1444
Stress Level	0.1437
Sleep Duration	0.1149
Heart Rate	0.0870
Age	0.0680
Steps_per_Hour_Sleep	0.0500
Activity_Stress_Index	0.0433
Daily Steps	0.0326
Gender_Male	0.0300

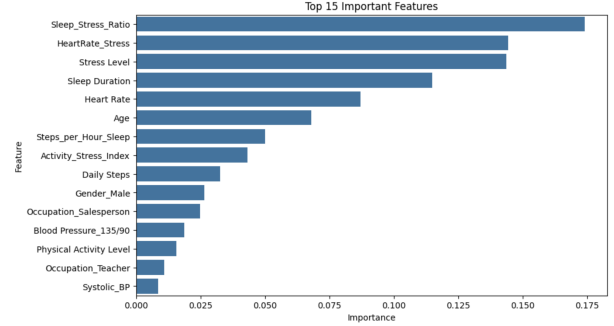


Fig. 6. Top 15 important features from Random Forest.

all 24 Good samples and all 23 Poor samples were classified correctly.

TABLE VIII
CONFUSION MATRIX OF BEST ADVANCED MODEL

Actual / Predicted	Good	Poor
Good	24	0
Poor	0	23

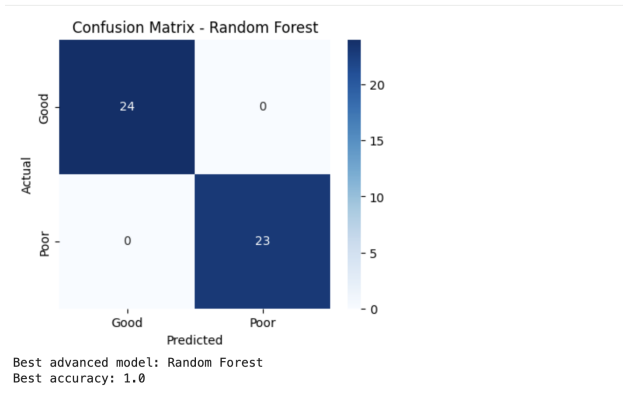


Fig. 7. Confusion matrix of the best advanced model.

XV. SHAP-BASED EXPLAINABILITY

The project also included SHAP analysis for model interpretability. SHAP stands for SHapley Additive exPlanations.

It explains how much each feature contributes to a model prediction. The SHAP value for a feature can be represented as:

$$\phi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(|N| - |S| - 1)!}{|N|!} [f(S \cup \{i\}) - f(S)] \quad (14)$$

SHAP helps explain why a sample is predicted as Good or Poor. In this project, SHAP was applied to the tuned Random Forest model. The SHAP summary plot can be used to identify global feature effects, while the force plot can explain individual predictions.



Fig. 8. SHAP summary plot for model interpretability.

XVI. RECOMMENDATION SYSTEM

A rule-based sleep improvement recommendation system was implemented. The system checks sleep duration, stress level, physical activity level, daily steps, and heart rate. Based on these values, it provides simple advice.

The recommendation rules include:

- If sleep duration is less than 7 hours, the system recommends sleeping at least 7 hours per night.
- If stress level is greater than or equal to 7, the system recommends meditation, breathing exercise, or screen-free time before sleep.
- If physical activity level is less than 40, the system recommends increasing daily physical activity.
- If daily steps are less than 6000, the system recommends walking more and aiming for 6000 to 8000 steps per day.
- If heart rate is greater than 80, the system recommends relaxation and professional consultation if needed.

This recommendation module makes the project more practical because it does not only predict sleep quality but also suggests possible improvements.

XVII. DISCUSSION

The results show that machine learning can effectively classify sleep quality using lifestyle and health data. The strong performance of Random Forest, Decision Tree, Logistic Regression, AdaBoost, Extra Trees, and Voting Classifier suggests that the dataset contains clear patterns between features and sleep quality labels.

However, very high accuracy should be interpreted carefully. The target variable was created from Quality of Sleep, and some engineered features may be strongly related to the original sleep quality score. Therefore, high performance may partly occur because the dataset contains strongly separable patterns. Cross-validation was useful for checking model stability. Logistic Regression achieved perfect cross-validation accuracy, while Random Forest and Hist Gradient Boosting also showed strong stability.

Feature importance analysis provided meaningful insights. Sleep_Stress_Ratio, HeartRate_Stress, Stress Level, Sleep Duration, and Heart Rate were the most influential features. These results are reasonable because stress and sleep duration directly affect sleep quality. Physical activity and daily steps also contributed to prediction, showing that lifestyle behavior is relevant for sleep health.

Clustering analysis also provided useful understanding. Cluster 0 had the best sleep behavior profile because it had longer sleep duration, lower stress, higher physical activity, and better sleep efficiency. Cluster 1 represented high-stress individuals with shorter sleep duration. Cluster 2 showed lower daily steps and higher heart rate. These cluster patterns can help identify groups of people who may need different types of sleep improvement support.

XVIII. LIMITATIONS

Although the project achieved strong results, there are some limitations. First, the dataset size is relatively small. A larger dataset would provide more reliable model evaluation. Second, the dataset may not represent all populations. Third, the classification target was generated from the original sleep quality score, so model performance depends heavily on this threshold-based label. Fourth, some models achieved perfect accuracy, which may indicate that the problem is highly separable or that the dataset contains features strongly correlated with the target.

XIX. FUTURE WORK

Future improvements can include collecting more diverse data from wearable devices, using real-time sleep monitoring, applying deep learning models, and adding more explainable AI methods. The recommendation system can also be improved by using medical guidelines and personalized user history. In addition, future research can use multi-class sleep quality prediction instead of only Good and Poor categories.

XX. CONCLUSION

This project presented a complete machine learning pipeline for sleep quality analysis. The work included preprocessing, clustering, classification, hyperparameter tuning, explainability, feature importance analysis, and recommendation generation. K-Means clustering identified three meaningful sleep behavior groups. In the classification task, Random Forest and several ensemble models achieved excellent performance. Advanced feature engineering improved model performance and helped identify important factors. The most influential features were related to stress, sleep duration, heart rate, and activity. Overall, the project demonstrates that machine learning can be an effective tool for sleep quality prediction and health-related decision support.

ACKNOWLEDGMENT

The author would like to thank the course instructor and department for guidance and support throughout this project.

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